

Brük, The open-source AI/ML platform for sovereign inference and training

Document status — draft / specification. Airon Brük is not yet built. This document describes the intended architecture, drawn from our services across several existing stacks we host. Everything is illustrative, not committed, and can be swapped during the project if a better-fitting alternative emerges. Treat this document as the shape of the stack.

What does Brük mean?

It reads as "brook." A bruk in Sweden, from the 1600s to the 1800s, was a vertically integrated iron community: ore mines, forges, housing, infrastructure, all owned and run by one operator.

About Airon

Airon designs, owns, and operates the full vertical — **datacenter, bare-metal machines, network fabric and software stack**. The first datacenter was built in Lidköping, Sweden, in 2022. Our niche is bare-metal: we deliver only **safe, performant, single-tenant** solutions to our customers. No GPU slicing, no shared tenancy, no shortcuts.

Brük is Airon's open-source platform that runs on top, completely optional for the user. It is fully infrastructure-agnostic and runs anywhere, on-premise or in any cloud.

Platform overview

What it is

Brük is an end-to-end, open-source platform for Inference-as-a-Service and Fine-Tuning-as-a-Service (tomorrow). The inference provides the end-user with an open-weight model library they can use with the provided OpenAI compatible endpoint.

Bare-metal and single-tenant by design

Each physical machine (node) is single-tenant, so the cluster carries no multi-tenant overhead. vLLM pods run inside **Kata Containers** for mutual host↔container isolation, GPUs are passed through whole, and cross-node KV transfer rides **NVIDIA Spectrum-X** (Ethernet RoCEv2).

Confidential compute support

Confidential compute (Confidential Containers (CoCo) + NVIDIA CC-mode GPU).

Purpose & objectives

This document is a project brief for the group taking on the build of Airon Bruk. The intent is to scope, evaluate and build an open-source, sovereign inference platform that:

- runs on bare-metal in single-tenant mode
- is Open source so that it can be deployed by anyone on their own environments or in any cloud
- supported confidential compute
- has a Agent operator managing the services

The group is expected to challenge every named component below, propose alternatives where they fit better and converge on a final stack that satisfies the objective. Treat this document as the *shape* of the system, not a fixed plan.

We will apply zero-trust principles at every layer of the stack, from software to hardware. Combining physical isolation with sandboxing gives us a strong foundation for tackling the classic confinement problem, including modern instances such as containing a rouge LLM.

Deliverables

By the end of the project the group should produce a running inference platform with an end-to-end security perspective, hosting an open-weight model library.

Infrastructure stack

#	Layer	Factory layer	What sits there in Airon Bruk
1	Datacenter / facility	Power, cooling, real estate	Our own data centers in Sweden
2	Machine	Server compute	Bare-metal node with H200, B200, B300 GPUs; AMD/Intel CPUs; BlueField-3 SuperNIC; BMC/IPMI on isolated management VLAN; public IP per node

3	Networking	Cluster fabric	NVIDIA Spectrum-X — Spectrum-4 Ethernet switches with adaptive routing and DOCA congestion control; BlueField-3 SuperNICs; RoCEv2 transport; NVLink for intra-node GPU peering
4	System software	Cluster management & OS	Hardened Ubuntu, vanilla Kubernetes, NVIDIA GPU Operator, Cilium CNI, Helm + Flux GitOps and Kata Containers as the default runtime (in non-CC)
5	AI platform	Inference and fine-tuning stack	NVIDIA Dynamo (smart router, NIXL, disaggregated serving), vLLM workers, Envoy etc.

In the project, every layer is hardened from physical to platform.

Design principles

Principle	What it means
OSI-approved licences only (<i>hard requirement</i>)	Every component shipped or depended on in the data path or control plane is licensed under an OSI-approved licence. Apache 2.0 is the preferred choice for the Airon-authored core; permissive (Apache, MIT, BSD, MPL) and weak-copyleft (LGPL) are accepted; AGPL is allowed only outside the request hot path;
Sovereign	In sovereign we also include the aspect that the user has access to the source code and can run the platform wherever, however, they want.
Bare-metal and single-tenant	Each bare-metal node belongs to one customer. No MIG slicing, no multi-tenant primitives at any level — the cluster <i>is</i> the tenant boundary.
Mutual host and container isolation	Every workload pod runs inside a Kata Containers . A vulnerability in a service cannot escape into the host kernel (the host kernel surface available to the workload is QEMU/KVM, not the full Linux kernel).
Defense in depth, bottom-up hardening	Every layer of the stack is hardened — physical (rack, ports, BMC), firmware, boot, OS, network, Kubernetes, workload sandbox, software supply chain and identity



Confidential

Confidential compute (Confidential Containers + NVIDIA CC compute is planned, mode with attestation)

possibly

post-evaluation

Reference architecture

[See attached Airon_bruk_reference_architecture.png]

Note over K8s,img: K8s schedules vLLM pods with runtimeClass kata-qemu. Kata isolates host from container in both directions via VM sandbox and VFIO GPU passthrough. Each node is single-tenant. No MIG. Cluster updates via Flux. Node OS via signed image.

Note over CC: TBD

Note over Obs: All layers async-stream spans, metrics, logs and audit events. Never on the request hot path.

Suggested components

Layer	Open-source / external components	Role
Front door	Cloudflare (SaaS) or customer's own DNS (on-prem / air-gapped)	Authoritative DNS — airon.computer for SaaS; customer's own zone for on-prem. Per-machine A records pointing to the node's public IP.
Edge / Gateway	Envoy, cert-manager	Terminates the mTLS, JWT/OIDC + OPA-WASM policy in-process, token-rate limiting, optional PII redaction, xDS routing into Dynamo.
Control plane	TBD	An Airon Operator agent will be deployed to manage the control-plane services. Specific component selections are still TBD.
Inference runtime	NVIDIA Dynamo, vLLM	KV-cache-aware smart routing, prefill/decode disaggregation, high-throughput LLM serving. Each

		pod runs in a Kata with GPU passthrough
Pod sandbox	Kata Containers by default, QEMU + KVM (kernel-side only), VFIO GPU passthrough	Mutual host and container isolation
Orchestration	Vanilla Kubernetes — upstream with Cilium CNI and NVIDIA GPU Operator, Helm + Flux for GitOps	Schedules pods onto the single-tenant nodes, runs the GPU driver/toolkit installer, eBPF networking.
Cluster fabric	NVIDIA Spectrum-X — Spectrum-4 Ethernet switches + BlueField-3 SuperNICs, RoCEv2 with adaptive routing and DOCA congestion control	Cross-node KV transfer for Dynamo's NIXL over standard RDMA.
Confidential compute	Confidential Containers (CoCo), Intel TDX, AMD SEV-SNP, NVIDIA CC-mode GPUs (H200, B200)	Planned, post-evaluation. Hardware-isolated workloads with four-part attestation; activated by runtimeClassName: kata-qemu-tdx on a confidential node pool.
Node OS & lifecycle	Hardened Ubuntu LTS, minimal package set, full Syft SBOM and Cosign signatures	The signed node image that Kubernetes runs on. Reproducible build; A/B image updates with auto-rollback.
Bare-metal substrate	NVMe local; shared storage with general compute (also bare-metal); public IP per node	Storage, fabric and addressability — the only layer customers must own on-prem.
Supply chain	Packer + Ansible (build-time only, not shipped), Syft, Cosign, in-toto / SLSA L3, Falco, Trivy	Reproducible image build, full SBOM, signed images, attestation evidence shipped with every release.
Observability	OpenTelemetry, Prometheus, Grafana	Traces, metrics, dashboards, audit events — all async, never on the request hot path.

Inference request flow

[See attached Airon_bruk_inference_request_flow.png]

1. **DNS** — client resolves e448dd9547db.airon.computer against DNS; record points at the public IP of a healthy bare-metal node w. per-node IP.
2. **Envoy** (host:443) — terminates the inner TLS, validates JWT and OPA policy, increments token-rate counter, applies PII redaction etc.
3. **Dynamo router** — KV-cache-aware routing picks the worker most likely to hit its prefix; disaggregates prefill (compute-bound pool) from decode (memory-bound pool). KV blocks move between them via NIXL over Spectrum-X (RoCEv2 Ethernet) cross-node, NVLink intra-node.
4. **vLLM worker** — runs the model inside a Kata microVM with whole-GPU passthrough. Additional alternative (branch): CoCo + NVIDIA CC-mode attestation handshake unwraps weights inside the TEE before serving.
5. **Streaming response** — back through Envoy to the client; OpenAI-compatible chunks for chat, gRPC streams for advanced clients.
6. **Telemetry** — OTel spans, Prometheus metering events, audit log, S3 Object Lock.

Hardening, bottom-up

Hardening is applied at every layer of the stack the platform owns, starting at the physical machine and working up. The list below is an example.

	Layer	Hardening posture
1	Physical	Locked racks; tamper-evident seals on chassis; cable port-level access controls; unused USB / serial / JTAG headers physically disabled or filled; BMC / IPMI on an isolated management VLAN with mTLS-only access
2	Firmware	Signed UEFI; Secure Boot enforced; vendor-signed BMC, NIC, GPU firmware only; TPM 2.0 measured boot extending into PCRO-7; unused devices and PCI ports disabled in BIOS / UEFI; IOMMU enabled
3	Boot chain	Measured boot (shim → grub → kernel → initrd) chained into TPM PCRs; UKI-signed kernel; dm-verity-protected immutable rootfs in the image; rollback only to a previously signed image
4	Node OS	Hardened Ubuntu LTS; CIS Level 2 baseline; FIPS-validated kernel via Ubuntu Pro; minimal package set; no compilers, build tools, or interactive shells reachable from the network; AppArmor and auditd in enforce mode

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| 5 | Network | SSH only via short-lived attested mTLS bastion; control plane on isolated VLAN; east-west via Cilium with default-deny NetworkPolicy; SPIFFE / SVID workload identity for service-to-service mTLS |
| 6 | Kubernetes | RBAC least-privilege; Pod Security Admission set to restricted; default-deny NetworkPolicies; no privileged pods, encrypted etcd |
| 7 | Workload sandbox | Kata microVM with VFIO GPU passthrough; read-only rootfs; dropped Linux capabilities; seccomp default profile; no host network; vsock-only IPC for kata-agent |
| 8 | Supply chain | Reproducible builds w. full Syft SBOM per release; Cosign signatures; SLSA L3 provenance; Trivy gate in CI; published OSI-licence inventory per release |
| 9 | Identity & secrets | OPA-WASM policy in-process in Envoy; Credentials bound to SPIFFE identity; KMS-wrapped secrets; HSM custody for production keys |
| 10 | Runtime detection | Falco rules for syscall-level anomalies; auditd on the host; Prometheus alerts on attestation failures, image-policy denials, and NetworkPolicy denials |
| 11 | Confidential (planned) | Adds: TDX / SEV-SNP CPU TEE encrypting RAM in use; NVIDIA CC-mode GPU (H200, B200) encrypting GPU HBM and SPDM-protecting PCIe; four-part Veraison attestation chain (TPM, CPU TEE, container digest, NVIDIA SPDM) gating any secret release |